UL Smart Campus Implementation Evidence and Presentation Command Book

SCOB032 Computer Networks Semester Project

Main Campus Polokwane and Satellite Campus

Field	Detail
Group	GROUP NUMBER 10
Packet Tracer file	UL-Smart-Campus-FINAL.pkt
Purpose	Presentation runbook, reconstructed feature configurations and verification commands

Security note: passwords, RADIUS keys and VPN pre-shared keys are intentionally redacted. This book records the implemented configuration, testing and troubleshooting evidence available from the project. The final Packet Tracer running configuration remains the authoritative device-level source.

1 How to Use This Book

This book is designed for the 10–12 minute live demonstration and for answering configuration questions from the assessors. The commands reconstruct the implemented design at feature level. They are not a replacement for configuration backups. Before presenting, compare every interface identifier and secret-free command with show running-config on the final Packet Tracer device.

Label	Meaning
Implemented	Confirmed from the completed Packet Tracer project and verification outputs.
Reconstructed	Command block recreated from the confirmed design; compare interface details with the final device.
Verification	Read-only command used during the live demonstration.
Do not run live	A configuration-changing command included for explanation only.

1.1 Presentation Safety Rules

- Open the final .pkt file before the presentation and do not make untested changes.
- Use show commands during the demonstration wherever possible.
- Keep a backup .pkt file in a separate folder.
- Do not display real passwords, RADIUS shared secrets or VPN pre-shared keys.
- Do not shut a link unless the HSRP failover demonstration has been rehearsed and can be restored quickly.

2 Addressing and VLAN Reference

VLAN	Name	Subnet	Gateway
10	ADMIN	10.10.7.0/25	10.10.7.1
20	STAFF	10.10.4.128/25	10.10.4.129
30	RESEARCH-LABS	10.10.7.128/25	10.10.7.129
35	PRINTERS	10.10.6.0/26	10.10.6.1
40	LIBRARY-USERS	10.10.4.0/25	10.10.4.1
50	RESIDENCE	10.10.8.0/22	10.10.8.1
60	SERVERS	10.10.6.192/26	10.10.6.193
70	VOICE	10.10.6.64/26	10.10.6.65
80	IOT-SECURITY	10.10.5.128/25	10.10.5.129
90	STAFF-WIFI	10.10.5.0/25	10.10.5.1
100	STUDENT-WIFI	10.10.0.0/22	10.10.0.1
110	MANAGEMENT	10.10.6.128/26	10.10.6.129
120	ICT-ADMIN	10.10.12.0/27	10.10.12.1

Satellite VLANs 20, 35, 40, 70, 80, 90, 100 and 110 use 10.20.[VLAN].0/24 with gateway 10.20.[VLAN].1.

3 Core Switching and HSRP

3.1 Main Core VLAN Creation

```
configure terminal
vlan 10
 name ADMIN
vlan 20
 name STAFF
vlan 30
 name RESEARCH-LABS
vlan 35
 name PRINTERS
vlan 40
 name LIBRARY-USERS
vlan 50
 name RESIDENCE
vlan 60
 name SERVERS
```

```
vlan 70
 name VOICE
vlan 80
 name IOT-SECURITY
vlan 90
 name STAFF-WIFI
vlan 100
 name STUDENT-WIFI
vlan 110
 name MANAGEMENT
vlan 120
 name ICT-ADMIN
ip routing
end
```

3.2 HSRP SVI Pattern

The following examples show the confirmed addressing pattern. CORE-SW1 uses priority 110 and preemption. CORE-SW2 uses priority 100 and preemption. Repeat the same pattern for every VLAN using the table above.

```
! CORE-SW1 example VLAN 100
interface Vlan100
 description STUDENT-WIFI-GATEWAY
 ip address 10.10.0.2 255.255.252.0
 ip helper-address 10.10.6.200
 ip access-group MAIN-STUDENT-ZERO-TRUST in
 standby 100 ip 10.10.0.1
 standby 100 priority 110
 standby 100 preempt
 no shutdown

! CORE-SW2 example VLAN 100
interface Vlan100
 description STUDENT-WIFI-GATEWAY
 ip address 10.10.0.3 255.255.252.0
 ip helper-address 10.10.6.200
 ip access-group MAIN-STUDENT-ZERO-TRUST in
 standby 100 ip 10.10.0.1
 standby 100 priority 100
 standby 100 preempt
 no shutdown
```

3.3 Confirmed Research SVI

```
! CORE-SW1
interface Vlan30
 description RESEARCH-LABS-GATEWAY
 ip address 10.10.7.130 255.255.255.128
 ip helper-address 10.10.6.200
 ip access-group RESEARCH-POLICY in
 standby 30 ip 10.10.7.129
 standby 30 priority 110
 standby 30 preempt

! CORE-SW2
interface Vlan30
 description RESEARCH-LABS-GATEWAY
 ip address 10.10.7.131 255.255.255.128
 ip helper-address 10.10.6.200
 ip access-group RESEARCH-POLICY in
 standby 30 ip 10.10.7.129
 standby 30 preempt
```

3.4 HSRP and STP Verification

```
show standby brief
show spanning-tree vlan 50
show spanning-tree vlan 90
show spanning-tree vlan 100
show interfaces trunk
show ip route
```

4 Access Switch Configuration Patterns

4.1 Redundant Core Uplinks

```
interface range GigabitEthernet0/1-2
 description REDUNDANT-CORE-UPLINK
 switchport mode trunk
 switchport trunk allowed vlan <SITE-VLAN-LIST>
 no shutdown
```

4.2 Secure User Access Port

```
interface FastEthernet0/<PORT>
 description <ENDPOINT-DESCRIPTION>
 switchport mode access
 switchport access vlan <DATA-VLAN>
 switchport port-security
 switchport port-security maximum <LIMIT>
 switchport port-security violation restrict
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown
```

4.3 Phone and PC Port

```
interface FastEthernet0/<PORT>
 description PHONE-AND-PC
 switchport mode access
 switchport access vlan <DATA-VLAN>
 switchport voice vlan 70
 spanning-tree portfast
 no shutdown
```

4.4 Unused Port Hardening

```
default interface FastEthernet0/<PORT>
interface FastEthernet0/<PORT>
 description UNUSED-SECURED-PORT
 shutdown
```

4.5 LAP Trunk Pattern

```
interface FastEthernet0/<LAP-PORT>
 description <BUILDING>-8021X-LAP
 no switchport port-security
 switchport mode trunk
 switchport trunk native vlan <LAP-NATIVE-VLAN>
 switchport trunk allowed vlan <USER-VLAN-LIST>
 spanning-tree portfast trunk
 no spanning-tree bpduguard enable
 no shutdown
```

Location	Switch port	Native VLAN	Allowed VLANs
ICT / Server room	MAIN-SERVER-SW Fa0/7	90	90,100
Main WLC	MAIN-SERVER-SW Fa0/8	90	50,90,100
Library	LIB-SW2 Fa0/5	90	90,100
Mathematics staff	MATH-STAFF-SW Fa0/7	90	90,100
Administration	ADMIN-SW1 Fa0/5	90	90,100
Lecture halls	LH-SW1 Fa0/2	90	90,100
M-Side residence	MSIDE-RES-SW Fa0/4-5	50	50,90
V-Side residence	VSIDE-RES-SW Fa0/4-5	Final .pkt value	50,90
Satellite LAP	SAT-SW1 Fa0/9	90	90,100
Satellite WLC	SAT-SW1 Fa0/10	90	90,100

5 Wired 802 1X and RADIUS

5.1 Research Laboratory Switches

```
aaa new-model
aaa authentication dot1x default group radius
radius-server host 10.10.6.201 auth-port 1645 key <REDACTED_RADIUS_KEY>
dot1x system-auth-control

interface range FastEthernet0/1-2
 description RESEARCH-LAB-COMPUTERS
 switchport mode access
 switchport access vlan 30
 authentication port-control auto
 dot1x pae authenticator
 spanning-tree portfast
 no shutdown

ip default-gateway 10.10.6.129
```

Management SVIs confirmed in the final project: MATH-LAB1-SW 10.10.6.145/26, MATH-LAB2-SW 10.10.6.146/26 and MATH-LAB3-SW 10.10.6.147/26.

5.2 Wireless Services

SSID	VLAN	Authentication	Purpose
UL-STAFF-8021X	90	RADIUS 802.1X	Staff wireless
UL-STUDENT-8021X	100	RADIUS 802.1X	Student wireless
UL-RESIDENCE-8021X	50	RADIUS 802.1X	Residence wireless

WLC configuration is completed through the Packet Tracer WLC interface, not IOS CLI. Present the WLAN list, RADIUS server, AP group, LAP status and connected client screens.

6 Zero Trust ACL Configuration

6.1 Student Policy

The final ACL must be copied from show access-lists MAIN-STUDENT-ZERO-TRUST. The confirmed functional order is required-service permits, protected-network denies, and Internet/default permit last.

```
ip access-list extended MAIN-STUDENT-ZERO-TRUST
 permit udp any any eq bootps
 permit udp any host 10.10.6.200 eq domain
 permit tcp any host 10.10.6.200 eq domain
 permit ip any 10.10.6.0 0.0.0.63
 permit ip any host 10.10.5.10
 deny ip any 10.10.0.0 0.0.255.255
 deny ip any 10.20.0.0 0.0.255.255
 permit ip any any
```

6.2 Research Policy

```
ip access-list extended RESEARCH-POLICY
 permit udp any host 10.10.6.200 eq domain
 permit tcp any host 10.10.6.200 eq domain
 permit ip any 10.10.6.0 0.0.0.63
 deny ip any 10.10.7.0 0.0.0.127
 deny ip any 10.10.4.128 0.0.0.127
 deny ip any 10.10.5.0 0.0.0.127
 deny ip any 10.10.5.128 0.0.0.127
 deny ip any 10.10.6.192 0.0.0.63
 deny ip any 10.10.6.64 0.0.0.63
 deny ip any 10.10.6.128 0.0.0.63
 permit ip any any
```

6.3 Server Policy

```
ip access-list extended SERVER-ZERO-TRUST
 permit ip 10.10.12.0 0.0.0.31 10.10.6.192 0.0.0.63
 permit udp any host 10.10.6.200 eq domain
 permit tcp any host 10.10.6.200 eq domain
 permit udp any host 10.10.6.200 eq bootps
 permit udp 10.10.6.128 0.0.0.63 host 10.10.6.201 eq 1645
 permit udp 10.10.6.128 0.0.0.63 host 10.10.6.201 eq 1812
 permit ip 10.10.5.128 0.0.0.127 host 10.10.6.202
 permit ip 10.20.80.0 0.0.0.255 host 10.10.6.202
 permit udp any host 10.10.6.203 eq 514
 permit udp any host 10.10.6.203 eq 123
 permit ip host 10.10.6.153 host 10.10.6.204
 permit udp host 10.10.5.10 host 10.10.6.201 eq 1645
 permit udp host 10.10.5.10 host 10.10.6.201 eq 1812
 permit udp host 10.20.90.10 host 10.10.6.201 eq 1645
 permit udp host 10.20.90.10 host 10.10.6.201 eq 1812
 deny ip any 10.10.6.192 0.0.0.63
 permit ip any any
```

6.4 ACL Verification

```
show access-lists MAIN-STUDENT-ZERO-TRUST
show access-lists RESEARCH-POLICY
show access-lists SERVER-ZERO-TRUST
```

7 Main and Satellite VPN

The following is the implemented design pattern. Confirm the exact crypto ACL name, transform-set name and sequence number from the final running configurations. Never display the real pre-shared key.

```
! MAIN-EDGE-RTR concept
crypto isakmp policy 10
 encr aes
 hash sha
 authentication pre-share
 group 2
crypto isakmp key <REDACTED_VPN_PSK> address 198.51.100.2
access-list <VPN-ACL> permit ip 10.10.0.0 0.0.255.255 10.20.0.0 0.0.255.255
crypto ipsec transform-set <TRANSFORM-SET> esp-aes esp-sha-hmac
crypto map UL-VPN 10 ipsec-isakmp
 set peer 198.51.100.2
 set transform-set <TRANSFORM-SET>
 match address <VPN-ACL>
interface FastEthernet1/0
 crypto map UL-VPN

! SAT-EDGE-RTR mirrors the configuration
! Local protected network 10.20.0.0/16
! Remote protected network 10.10.0.0/16
! Peer 203.0.113.1
! Crypto map applied to FastEthernet0/1
```

7.1 VPN Verification

```
show crypto isakmp sa
show crypto ipsec sa
```

Presentation evidence: QM_IDLE, ACTIVE inbound/outbound SAs, and increasing encapsulation/decapsulation counters.

8 Voice Configuration

8.1 Main Voice Router

```
telephony-service
 max-ephones 20
 max-dn 20
 ip source-address 10.10.6.68 port 2000
```

```
 auto assign 1 to 20

ephone-dn 1
 number 2001
ephone-dn 2
 number 2002
ephone-dn 3
 number 2003
ephone-dn 4
 number 2004
ephone-dn 5
 number 2005
ephone-dn 6
 number 2006
ephone-dn 7
 number 2007
ephone-dn 8
 number 2008
ephone-dn 9
 number 2009

dial-peer voice 300 voip
 destination-pattern 3...
 session target ipv4:10.20.70.2
```

8.2 Satellite Voice Router

```
! Local extension confirmed as 3001
dial-peer voice 200 voip
 destination-pattern 2...
 session target ipv4:10.10.6.68
```

8.3 Voice Verification

```
show ephone
show running-config | section telephony-service
show running-config | section ephone-dn
show running-config | section dial-peer
```

Live demonstration: place a call between 3001 and 2004, answer it and show Connected.

9 Quality of Service

9.1 Main Edge QoS

```
ip access-list extended MAIN-VOICE-QOS
 permit ip 10.10.6.64 0.0.0.63 any

class-map match-any MAIN-VOICE-CLASS
 match access-group name MAIN-VOICE-QOS
 match ip dscp ef

policy-map MAIN-VOICE-MARK
 class MAIN-VOICE-CLASS
  set dscp ef

policy-map MAIN-WAN-QOS
 class MAIN-VOICE-CLASS
  priority percent 20
 class MAIN-VIDEO-CLASS
  bandwidth percent 30
 class class-default
  fair-queue

interface range FastEthernet0/0-1
 service-policy input MAIN-VOICE-MARK
interface FastEthernet1/0
 service-policy output MAIN-WAN-QOS
```

9.2 Satellite Edge QoS

```
ip access-list extended SAT-VOICE-QOS
 permit ip 10.20.70.0 0.0.0.255 any

class-map match-any SAT-VOICE-CLASS
 match access-group name SAT-VOICE-QOS
 match ip dscp ef

policy-map SAT-VOICE-MARK
 class SAT-VOICE-CLASS
  set dscp ef

policy-map SAT-WAN-QOS
 class SAT-VOICE-CLASS
  priority percent 20
 class class-default
  fair-queue

interface FastEthernet0/0
 service-policy input SAT-VOICE-MARK
interface FastEthernet0/1
 service-policy output SAT-WAN-QOS
```

9.3 QoS Verification

```
! MAIN-EDGE-RTR
show policy-map interface FastEthernet0/0
show policy-map interface FastEthernet0/1
show policy-map interface FastEthernet1/0

! SAT-EDGE-RTR
show policy-map interface FastEthernet0/0
show policy-map interface FastEthernet0/1
```

Packet Tracer limitation: qos pre-classify is rejected under the crypto map, and encrypted RTP/DSCP counters are not fully reproduced. Inside policies still prove voice identification and marking; WAN policies prove the strict-priority design is attached.

10 Management Monitoring and Services

```
! NTP and syslog pattern
ntp server 10.10.6.203
logging host 10.10.6.203

! SSH pattern - confirm final domain and local username
ip domain-name ul.ac.za
crypto key generate rsa
ip ssh version 2
line vty 0 4
 login local
 transport input ssh
```

Service	Address	Presentation evidence
DNS and DHCP	10.10.6.200	DNS records, DHCP pools and client ipconfig
RADIUS	10.10.6.201	Network clients and users without passwords
IoT server	10.10.6.202	Registered cameras and sensors
NTP and syslog	10.10.6.203	Clock and received logs
SDN controller	10.10.6.204	Managed device inventory
Main WLC	10.10.5.10	WLANs, AP groups, LAPs and clients

11 SDN Monitoring Script

```
import json
import os
import requests

CONTROLLER = "http://10.10.6.204"
```

```
USERNAME = os.getenv("SDN_USERNAME", "sdnadmin")
PASSWORD = os.getenv("SDN_PASSWORD")

if not PASSWORD:
    raise SystemExit("SDN_PASSWORD is not configured.")

ticket_url = CONTROLLER + "/api/v1/ticket"
devices_url = CONTROLLER + "/api/v1/network-device"

login_response = requests.post(
    ticket_url,
    data=json.dumps({"username": USERNAME, "password": PASSWORD}),
    headers={"content-type": "application/json"},
    timeout=30
)

print("Authentication status:", login_response.status_code)
if login_response.status_code not in (200, 201):
    raise SystemExit("Authentication failed.")

ticket = login_response.json()["response"]["serviceTicket"]
device_response = requests.get(
    devices_url,
    headers={"x-auth-token": ticket, "content-type": "application/json"},
    timeout=30
)

print("Device request status:", device_response.status_code)
if device_response.status_code != 200:
    raise SystemExit("Could not retrieve managed devices.")

devices = device_response.json()["response"]
print("Managed devices found:", len(devices))

for device in devices:
    print("Hostname:", device.get("hostname", "Unknown"))
    print("Management IP:", device.get("managementIpAddress", "Unknown"))
    print("Device type:", device.get("type", "Unknown"))
    print("Reachability:", device.get("reachabilityStatus", "Unknown"))
    print("Collection status:", device.get("collectionStatus", "Unknown"))
    print("--------------------------------")
```

12 Simulated Security Attacks

Test	Action	Expected evidence
Student lateral movement	Student pings ICT Admin or SDN server	Failure plus increased deny counter in MAIN-STUDENT-ZERO-TRUST
Invalid 802.1X login	Use incorrect wireless credentials	Authentication denied and no valid address
Rogue wired endpoint	Connect different device to a secured access port	Port-security violation or restricted traffic
IoT lateral movement	IoT tests authorised server then protected network	IoT server permitted; protected destination denied
Unauthorised SSH	Student attempts device management	Denied; ICT Admin succeeds

13 Ten to Twelve Minute Demonstration Script

Time	Presenter action	Command or screen	What to say
0:00–1:00	Introduce scenario and objectives	Complete topology	Two sites, 500+ users, segmented wired/wireless, voice and IoT.
1:00–2:00	Explain VLSM and VLANs	Addressing diagram	Address space follows demand; /22 for large pools and smaller subnets for services.
2:00–3:15	Show redundancy	show standby brief; show spanning-tree vlan 100	HSRP provides gateway failover and STP prevents loops across redundant uplinks.
3:15–4:30	Show zero-trust security	show access-lists MAIN-STUDENT-ZERO-TRUST	Required services are permitted while lateral movement is denied.

4:30–5:30	Show VPN	show crypto isakmp sa; show crypto ipsec sa	QM_IDLE and active SAs prove encrypted inter-campus traffic.
5:30–7:00	Show wireless 802.1X	WLC LAP/client screens	RADIUS authenticates staff, students and residence users into separate VLANs.
7:00–8:00	Show VoIP	Phone 3001 calls 2004; show ephone	Local CME and dial peers support working inter-campus calls.
8:00–9:00	Show QoS	show policy-map interface ...	Voice is identified, marked EF and assigned 20% strict priority.
9:00–10:00	Show IoT and SDN	IoT browser and SDN script output	IoT is isolated, while the SDN controller provides central visibility.
10:00–11:00	Explain testing and troubleshooting	Evidence screenshots	DHCP, ACLs, HSRP, STP, VPN, authentication and services were tested.
11:00–12:00	Conclude	Recommendations slide	The design is secure, resilient and budget-conscious; production needs EAP-TLS, SIEM and physical validation.

14 Fast Verification Checklist

- Both core switches show the expected HSRP Active and Standby roles.
- VPN shows QM_IDLE and active security associations.
- All required LAPs are online.
- Representative wired and wireless clients receive correct DHCP addresses.
- Research wired 802.1X and wireless 802.1X succeed with valid credentials.
- Student, research and IoT ACL counters demonstrate policy enforcement.
- Main and satellite phones are registered.
- The 3001 to 2004 call displays Connected.
- QoS marking and WAN policies are attached.
- IoT devices are visible on the IoT server.
- The SDN controller and monitoring script return device inventory.
- NTP, syslog and SSH evidence is ready.
- Final and backup Packet Tracer files are saved.

Appendix A Configuration Backup Commands

```
enable
show running-config
show startup-config
show ip interface brief
show interfaces status
show interfaces trunk
show vlan brief
show spanning-tree
show standby brief
show ip route
show access-lists
show cdp neighbors
write memory
```

Export show running-config from every router and switch. Use screenshots or configuration records for WLCs and Packet Tracer servers. The actual exported running configurations must accompany this reconstructed presentation book.

Appendix B Project Architecture Diagrams

These diagrams consolidate the implemented physical and logical design, VLAN allocation, security architecture and SDN or monitoring relationships. Use them when explaining the topology before opening individual device configurations.

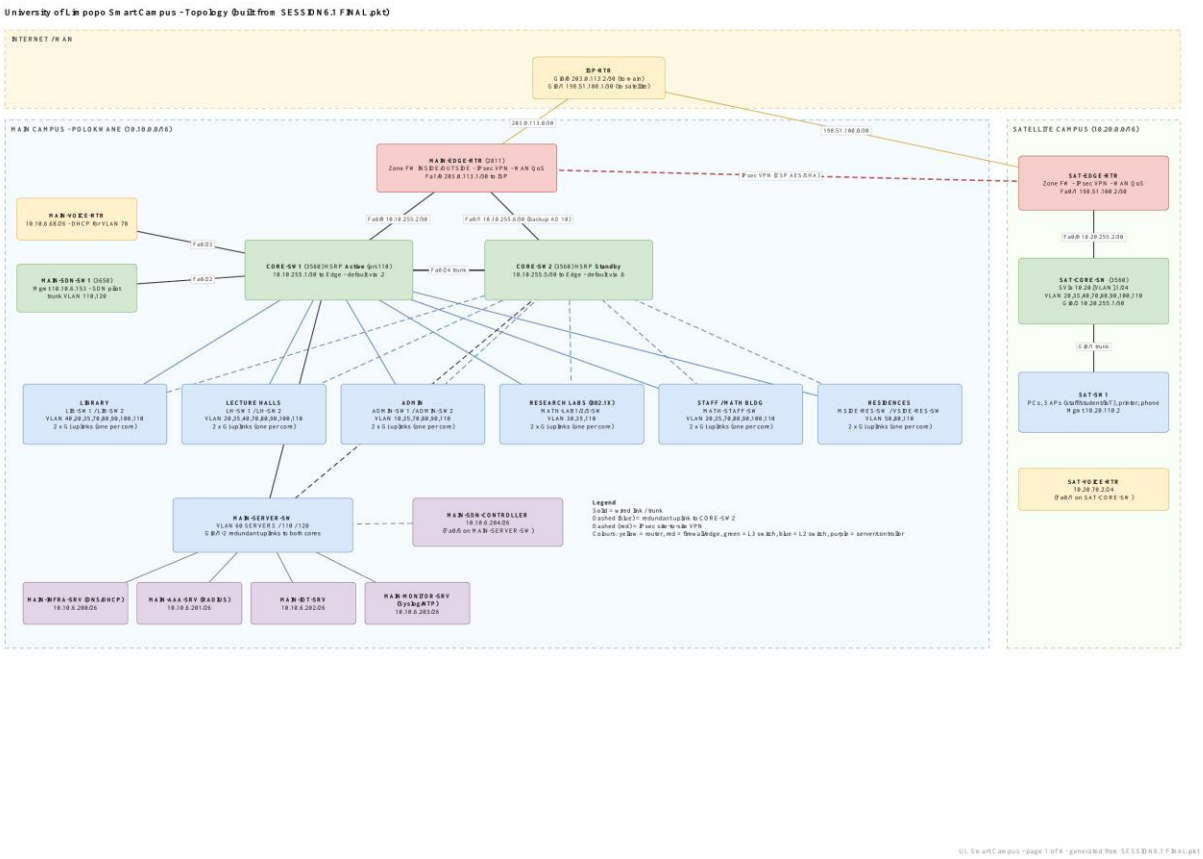


Figure B1 Physical and logical multi-site campus topology

VLAN and IP addressing plan – Main campus (10.10.0.0/16) and Satellite (10.20.0.0/16)

VLAN	Name	Subnet	Mask	HSRP VIP (gw)	Notes
10	ADMIN	10.10.7.0	/25	10.10.7.1	
20	STAFF	10.10.4.128	/25	10.10.4.129	Protected from student VLANs
30	RESEARCH-LABS	10.10.7.128	/25	10.10.7.129	RESEARCH-POLICY in - 802.1X
35	PRINTERS	10.10.6.0	/26	10.10.6.1	
40	LIBRARY-USERS	10.10.4.0	/25	10.10.4.1	MAIN-STUDENT-ZERO-TRUST in
50	RESIDENCE	10.10.8.0	/22	10.10.8.1	MAIN-STUDENT-ZERO-TRUST in
60	SERVERS	10.10.6.192	/26	10.10.6.193	SERVER-ZERO-TRUST out
70	VOICE	10.10.6.64	/26	10.10.6.65	Voice VLAN - QoS
80	IOT-SECURITY	10.10.5.128	/25	10.10.5.129	MAIN-IOT-ZERO-TRUST in
90	STAFF-WIFI	10.10.5.0	/25	10.10.5.1	
100	STUDENT-WIFI	10.10.0.0	/22	10.10.0.1	MAIN-STUDENT-ZERO-TRUST in
110	MANAGEMENT	10.10.6.128	/26	10.10.6.129	Switch mgmt .140–.153
120	ICT-ADMIN	10.10.12.0	/27	10.10.12.1	ICT-MANAGEMENT-ONLY
P2P	Core-Edge links	10.10.255.0	/30 x2	.1/.2 and .5/.6	Fa0/0 and Fa0/1 on edge
WAN	Edge-ISP	203.0.113.0	/30	.1 / .2	Main side
WAN	Sat-ISP	198.51.100.0	/30	.1 / .2	Satellite side
SAT	VLAN 20,35,40,70,80,90,100,110	10.20.[VLAN].0	/24 each	10.20.[VLAN].1	Gateway = SAT-CORE-SW SVI; Satcore link 10.20.255.0/30

VLSM: /22 for the largest user pools (Residence 1022 hosts, Student WiFi 1022 hosts), /25 for building VLANs, /26 for services and management, /27 for ICT admin, /30 for point-to-point links. Each gateway is an HSRP virtual IP shared by CORE-SW1 (x2/x3 pattern per VLAN) and CORE-SW2.

UL Smart Campus - page 2 of 4 - generated from SESSION6.1 FINAL.pkt

Figure B2 Main and satellite VLAN and addressing plan

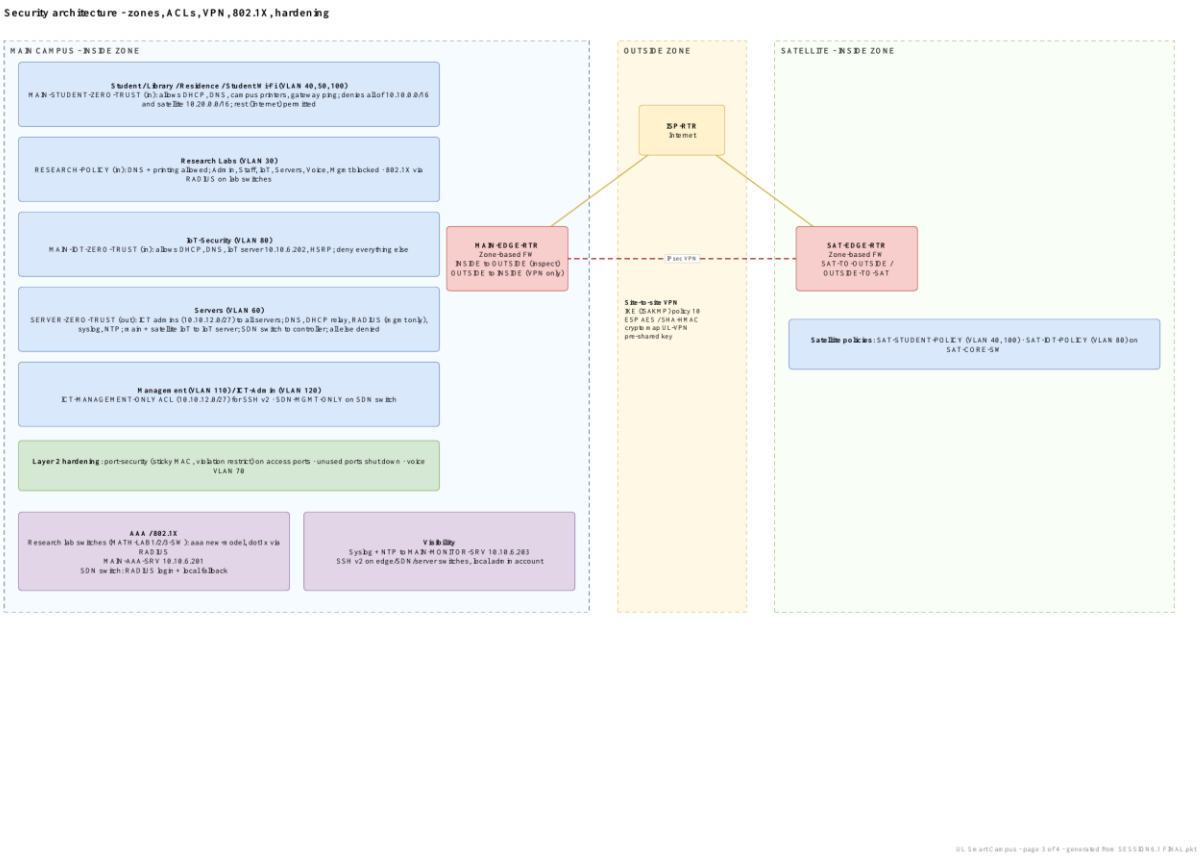


Figure B3 Security architecture including ACLs, 802.1X, VPN and management controls

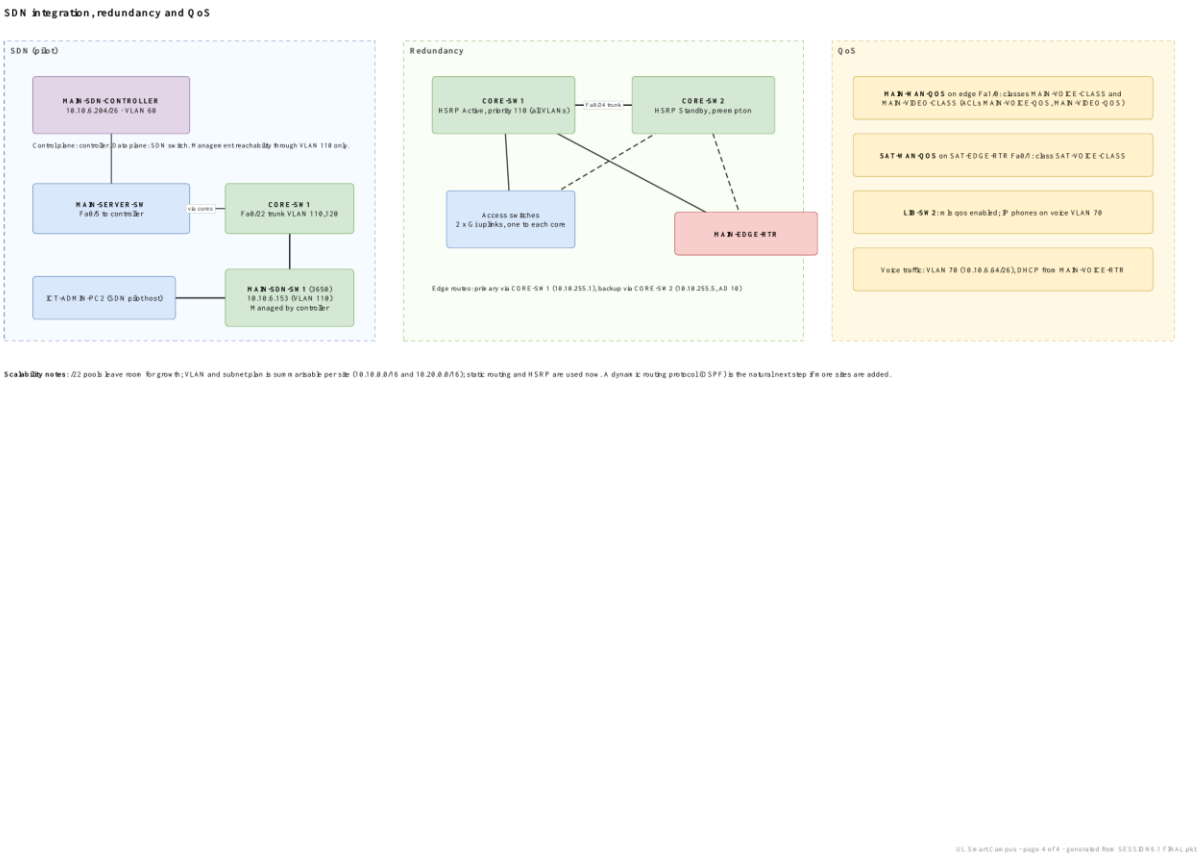


Figure B4 SDN integration, monitoring and IoT service relationships

Appendix C Implemented Device and Feature Register

Area	Devices or components	Implemented function
Main core	CORE-SW1 and CORE-SW2	SVIs, inter-VLAN routing, HSRP, DHCP relay, ACL attachment, STP root control and redundant trunks
Main WAN edge	MAIN-EDGE-RTR	Static WAN routing, IPsec VPN termination, zone-policy inspection and WAN QoS
Satellite core	SAT-CORE-SW	Satellite SVIs, gateways, routing, DHCP relay and student policy attachment
Satellite WAN edge	SAT-EDGE-RTR	WAN routing, IPsec VPN termination, security policy and WAN QoS
Voice	MAIN-VOICE-RTR and SAT-VOICE-RTR	CME registration, extensions, dial peers and inter-campus calling
Wireless	MAIN-WLC1, SAT-WLC1 and all LAPs	Staff, student and residence WLANs, RADIUS, AP groups, VLAN mapping and client access
Access layer	Admin, Library, Lecture Hall, Mathematics, Residence, Server and Satellite switches	Access VLANs, voice VLANs, LAP and WLC trunks, dual core uplinks, port security, PortFast, BPDU Guard and unused-port shutdown
Central services	10.10.6.200 to 10.10.6.204	DHCP and DNS, RADIUS, IoT, NTP and syslog, and SDN controller services
Automation	SDN monitoring Python script	Controller ticket authentication and managed-device inventory retrieval

Appendix D Comprehensive Test Results

Test	Method or evidence	Observed result	Status
VLAN and trunk propagation	show vlan brief and show interfaces trunk	Required VLANs active on LAP, WLC, access and core paths	Passed
DHCP allocation	Client DHCP selection, ipconfig and helper-address verification	Representative wired and wireless clients received the correct subnet, gateway and DNS values after convergence	Passed
Inter-VLAN routing	Gateway and authorised-service pings	Permitted traffic routed through the active SVI gateway	Passed
Student isolation	Ping protected networks and inspect MAIN-STUDENT-ZERO-TRUST counters	Protected destinations denied after required DNS, DHCP and service permits	Passed
Research isolation	Test VLAN 30 service access and protected destinations	RESEARCH-POLICY allowed authorised servers and denied unrelated segments	Passed
Server protection	Access central services from approved and unapproved networks	SERVER-ZERO-TRUST counters confirmed permits and denials	Passed
HSRP failover	Interrupt primary gateway path and run show standby brief	CORE-SW2 became Active; CORE-SW1 resumed Active after restoration and preemption	Passed
STP redundancy	show spanning-tree vlan 50, 90 and 100	Primary uplinks forwarded and redundant links blocked where required	Passed
VPN	show crypto isakmp sa and show crypto ipsec sa	QM_IDLE, active security associations and increasing encryption counters	Passed
Wired 802.1X	Research PC with valid and invalid RADIUS credentials	Valid client authenticated after endpoint 802.1X was enabled	Passed
Wireless 802.1X	Staff, student and residence clients joined mapped SSIDs	Clients received addresses from VLAN 90, 100 or 50 respectively	Passed
LAP operation	WLC AP-group status and switch MAC learning	Required LAPs reported online and learned management MAC addresses	Passed
VoIP registration	show ephone on both voice routers	Main extensions 2001 to 2009 and satellite extension 3001 registered	Passed
Inter-campus call	Call from 3001 to 2004	Both phones displayed Connected	Passed

QoS classification	show policy-map interface on inside and WAN interfaces	Inside voice policy matched traffic; WAN priority policies remained attached	Passed with simulator limitation
IoT registration	ICT Admin browser to the IoT server	Registered cameras and sensors visible through the browser service	Passed
SDN monitoring	Controller browser and Python API request	Controller reachable and managed-device inventory returned	Passed
NTP and syslog	show clock, NTP configuration and server log view	Clock synchronisation and central log collection configured	Passed
SSH management	Ping and SSH from the authorised management endpoint	Reachability confirmed; verify final SSH service on every device	Final verification required

Appendix E Troubleshooting and Corrective Actions

Problem observed	Cause identified	Corrective action	Verification
VLAN 100 absent from an uplink path	Student VLAN was not allowed or created locally	Created VLAN 100 and added it to both ends of the required trunks	Allowed and active lists included VLAN 100; STP reached Forwarding
An access switch became STP root for a new VLAN	Core priority had not been set for that VLAN	Set CORE-SW1 priority 24576 and CORE-SW2 priority 28672	CORE-SW1 displayed This bridge is the root
Core trunk mode command rejected	Trunk encapsulation remained Auto	Selected dot1q where supported, then configured trunk mode and allowed VLANs	show interfaces trunk displayed on and 802.1Q
Residence LAP stayed offline or joined slowly	Native VLAN, DHCP convergence or stale simulator state	Aligned native VLAN and allowed VLANs, bounced the port only when needed and allowed convergence time	MAC learned on the native VLAN and WLC status changed to Online
Wireless client showed no IPv4 address	Profile had not re-authenticated or DHCP had not completed	Reselected the SSID, confirmed 802.1X and selected DHCP after association	Client received an address from the mapped VLAN
Research wired 802.1X failed	Endpoint 802.1X security option was disabled	Enabled endpoint 802.1X and retried valid credentials	Authentication and DHCP succeeded
Student ACL did not isolate traffic	A broad permit ip any any preceded the deny statements	Removed the misplaced permit and re-added it after the internal denies	ACL order showed specific permits, denies and the final Internet permit
HSRP temporarily showed both cores Active	VLAN path or Packet Tracer convergence interrupted HSRP hellos	Restored the trunk path and allowed hello convergence	CORE-SW1 Active and CORE-SW2 Standby were restored
QoS WAN voice counters stayed near zero	Packet Tracer encrypts before the WAN classifier and rejects qos pre-classify	Classified and marked voice on inside interfaces and matched DSCP EF at the WAN edge	Inside counters increased, WAN policy was attached and the call succeeded
SSH returned connection refused	Target was reachable but SSH or VTY configuration was incomplete	Verify RSA keys, local user, SSH version 2 and VTY login local	Re-test from ICT Admin and capture the accepted session
Obsolete AP ports remained configured	Autonomous APs had been replaced by controller-based LAPs	Defaulted, labelled and shut the old AP interfaces	show interfaces status displayed disabled VLAN 1
Invalid CLI input appeared	Commands were prefixed with how, sshow or the device prompt	Re-entered only the IOS command after the prompt	Correct show output returned without changing configuration

Appendix F Screenshot Evidence

The screenshots below are representative evidence captured during implementation and troubleshooting. CLI outputs in the main body remain the clearest evidence for HSRP, VPN, ACL, trunk and QoS state.

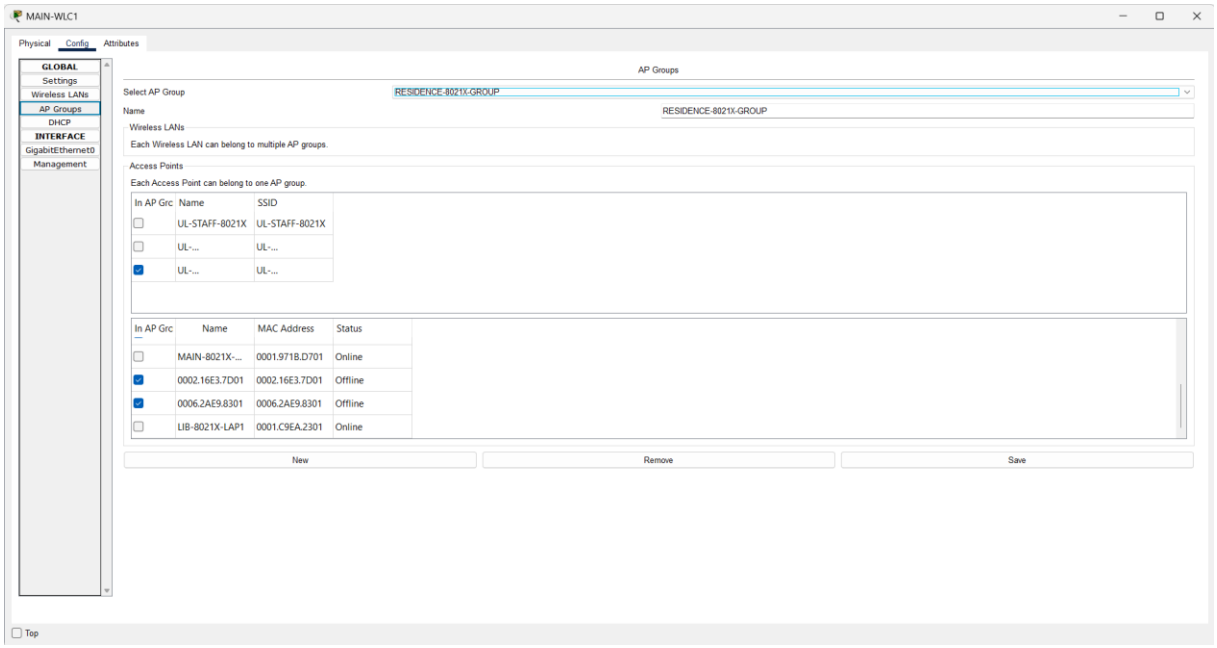
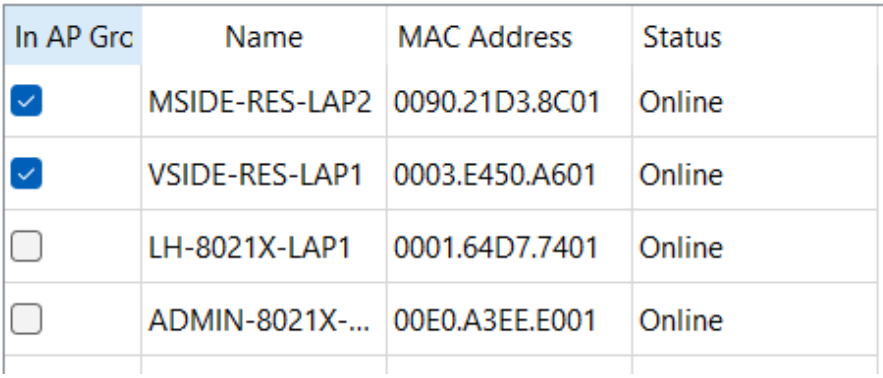
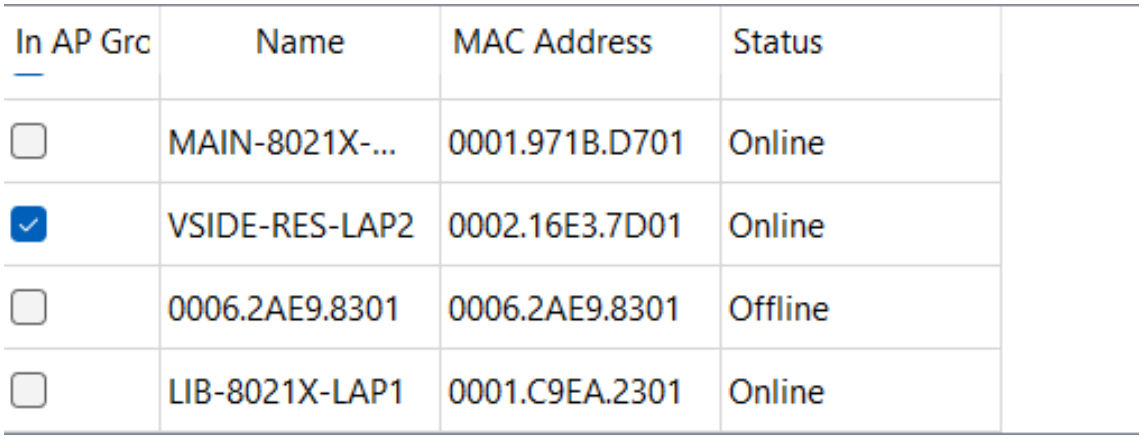


Figure F1 Main WLC lightweight access point inventory during residence troubleshooting



In AP Grc	Name	MAC Address	Status
☑	MSIDE-RES-LAP2	0090.21D3.8C01	Online
☑	VSIDE-RES-LAP1	0003.E450.A601	Online
☐	LH-8021X-LAP1	0001.64D7.7401	Online
☐	ADMIN-8021X-...	00E0.A3EE.E001	Online

Figure F2 AP group membership showing residence and campus LAP status



In AP Grc	Name	MAC Address	Status
☐	MAIN-8021X-...	0001.971B.D701	Online
☑	VSIDE-RES-LAP2	0002.16E3.7D01	Online
☐	0006.2AE9.8301	0006.2AE9.8301	Offline
☐	LIB-8021X-LAP1	0001.C9EA.2301	Online

Figure F3 AP group membership exposing an offline LAP before corrective action

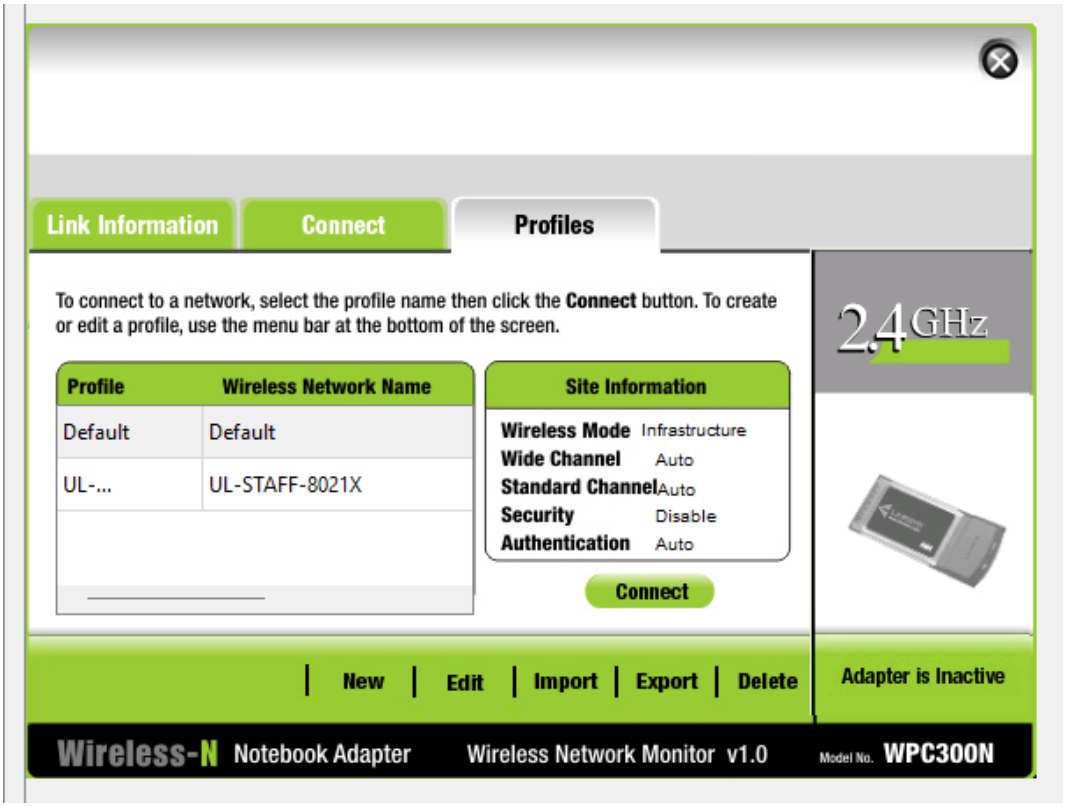


Figure F4 Wireless client profile with SSID and 802.1X settings

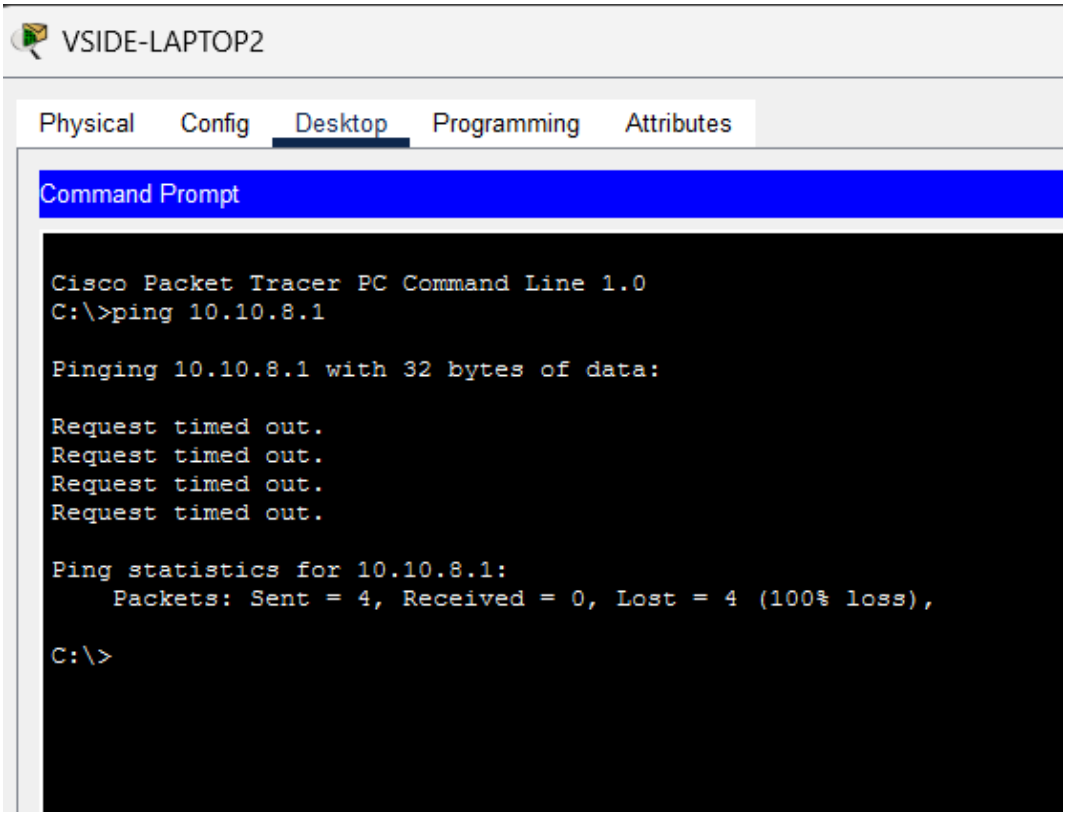


Figure F5 Failed reachability test used during client troubleshooting

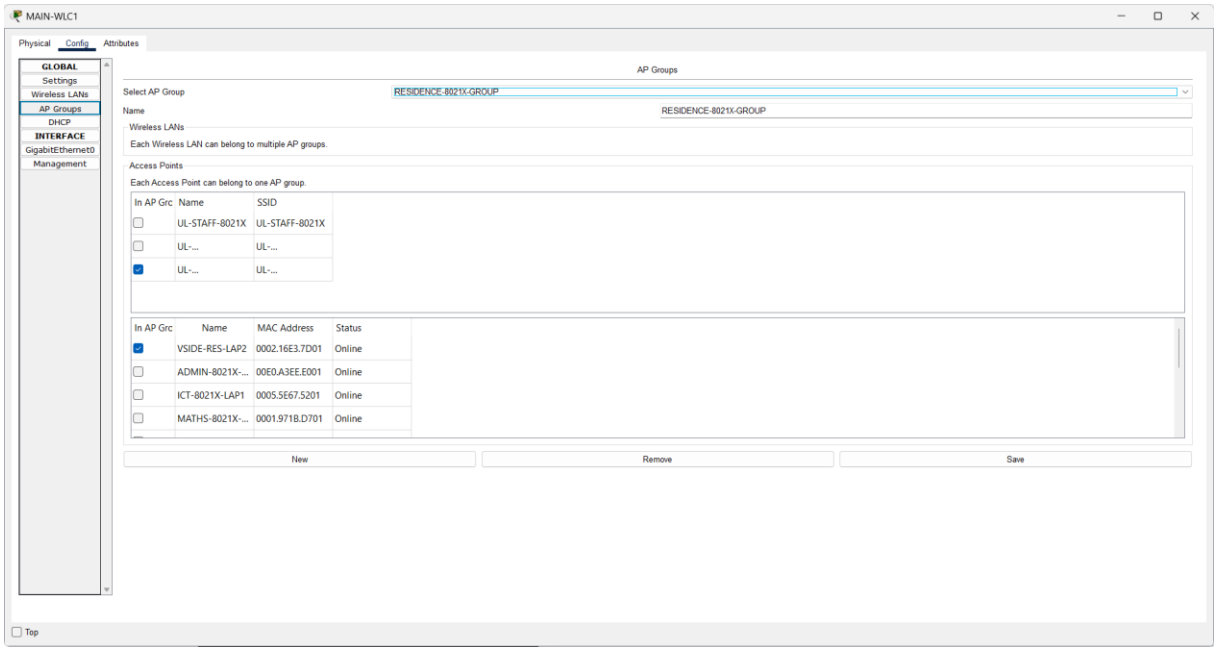


Figure F6 WLC client or WLAN verification after corrective configuration

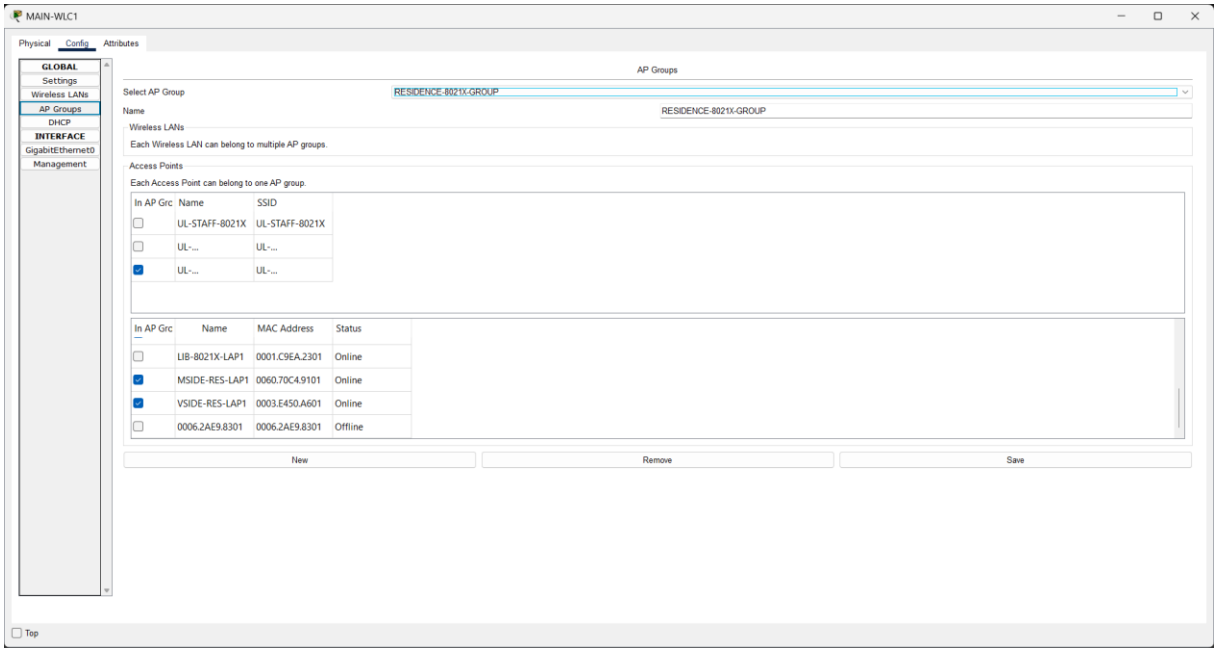


Figure F7 Wireless controller status evidence

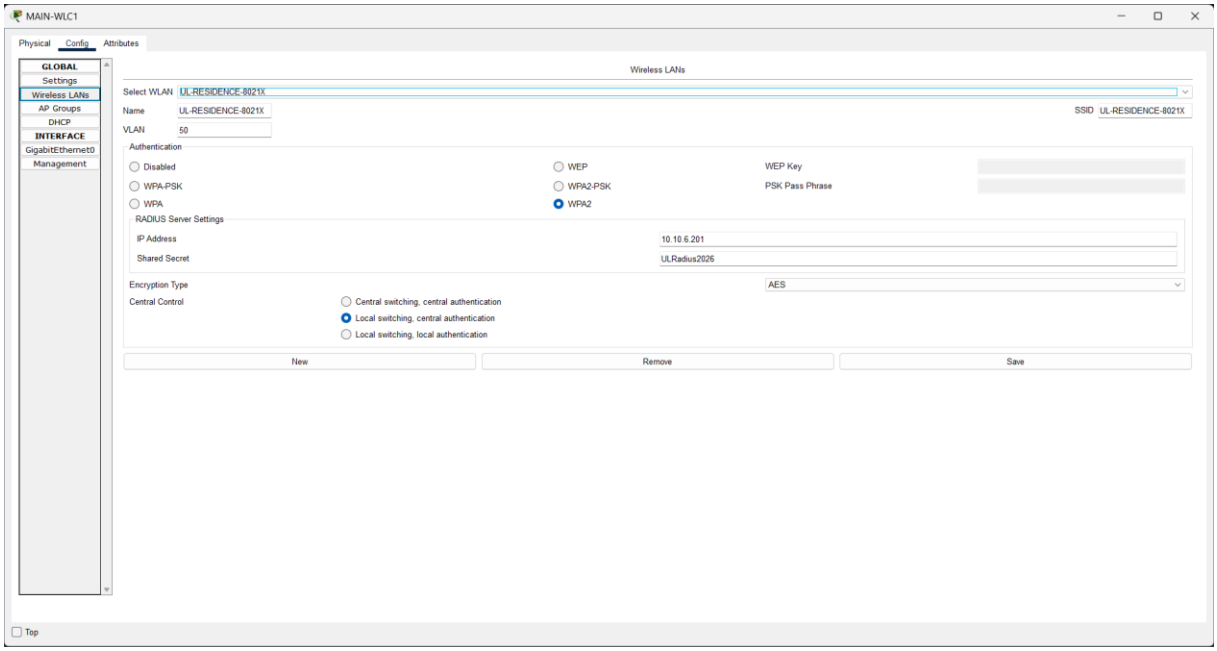


Figure F8 Wireless or RADIUS configuration evidence

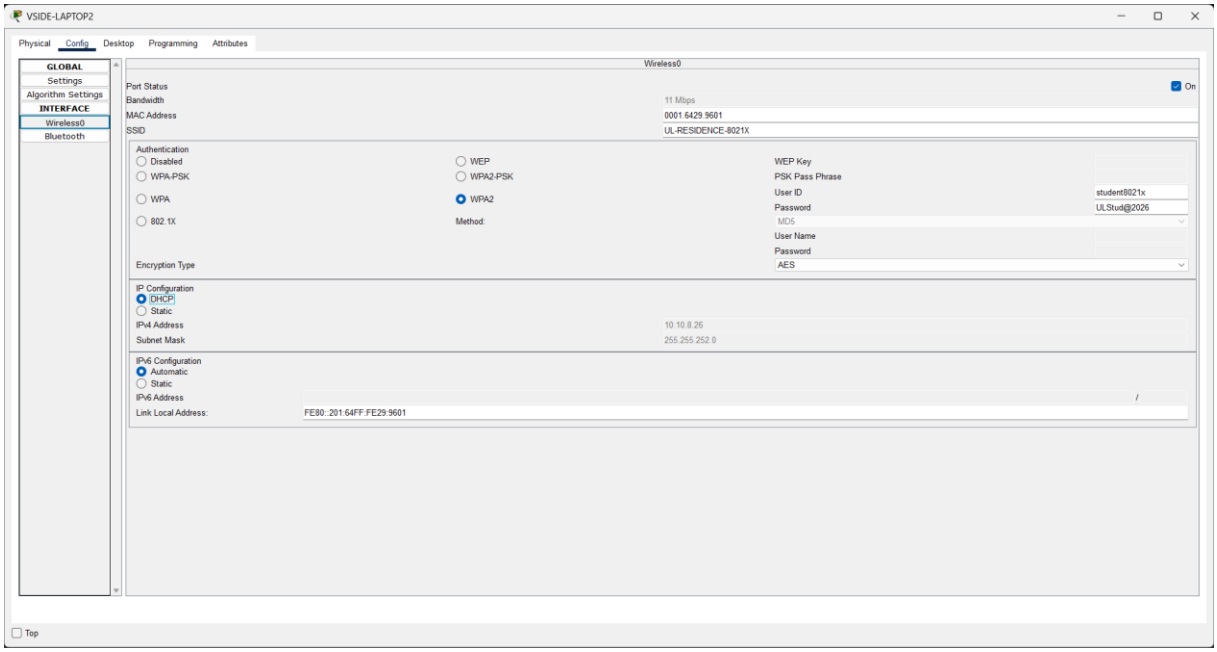


Figure F9 Final wireless service configuration evidence



Figure F10 Inter-campus phone call showing the connected call state

Appendix G Final Evidence Still to Capture

These screenshots cannot be recreated reliably from text alone. Capture them from the final saved Packet Tracer file immediately before submission. Remove an item only after its evidence is stored with the project.

- Complete Packet Tracer topology at readable zoom, plus one close-up per campus.
- CORE-SW1 and CORE-SW2 show standby brief with stable Active and Standby roles.
- Representative show spanning-tree output for VLANs 50, 90 and 100.
- Both show crypto isakmp sa and show crypto ipsec sa outputs.
- Final WLC WLAN list, AP groups, all required LAPs online and connected-client list.
- Valid and invalid wired and wireless 802.1X tests without exposing passwords.
- DHCP pool screens and client ipconfig for main, residence and satellite networks.
- ACL output before and after a denied student or IoT lateral-movement test.
- Main and satellite show ephone outputs and the 3001 to 2004 connected call.
- Inside and WAN show policy-map interface outputs for QoS.
- IoT registered-device page, SDN inventory and Python script output.
- NTP clock, syslog events and one successful authorised SSH session.